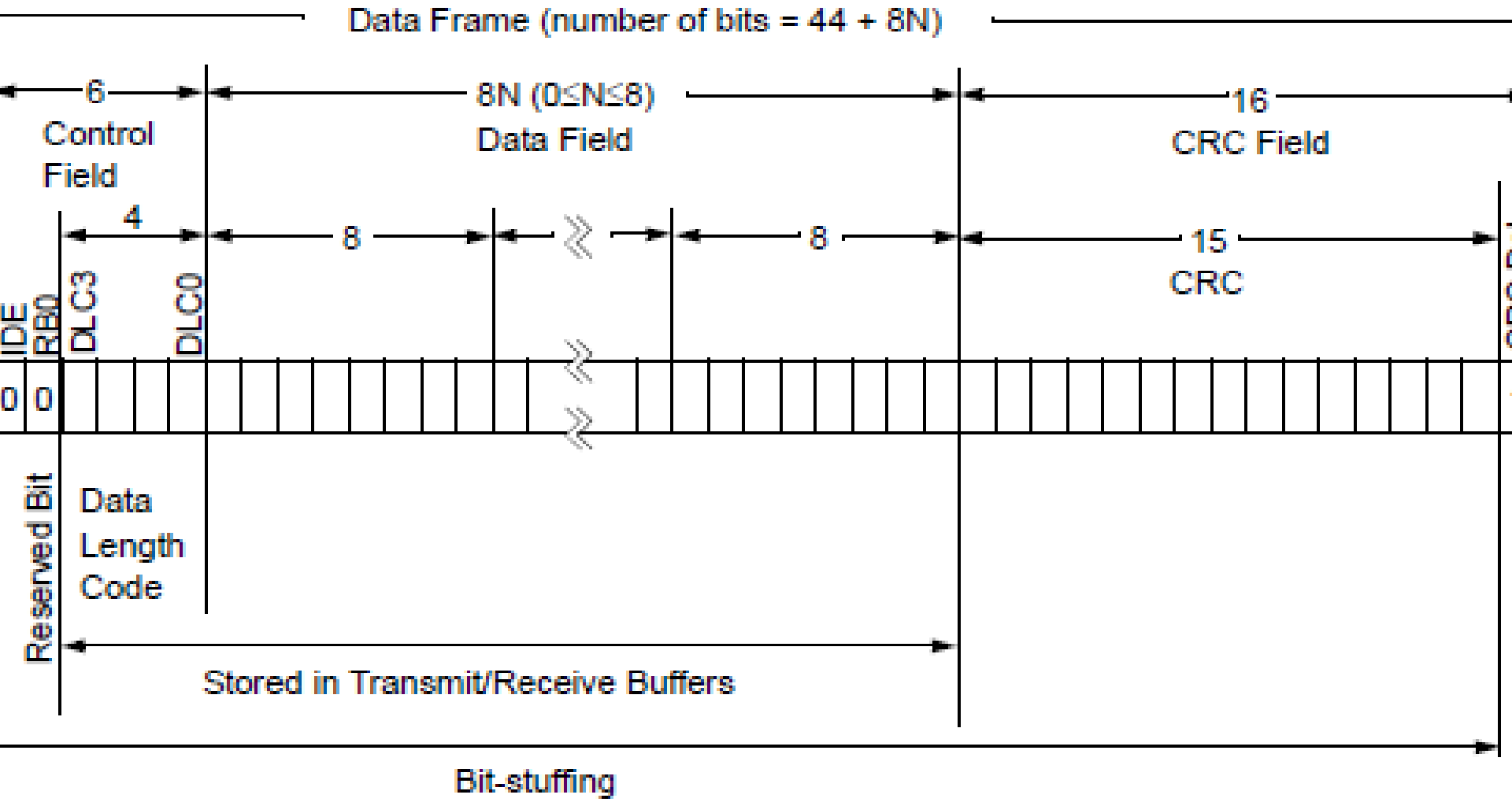
CAN PROTOCOL

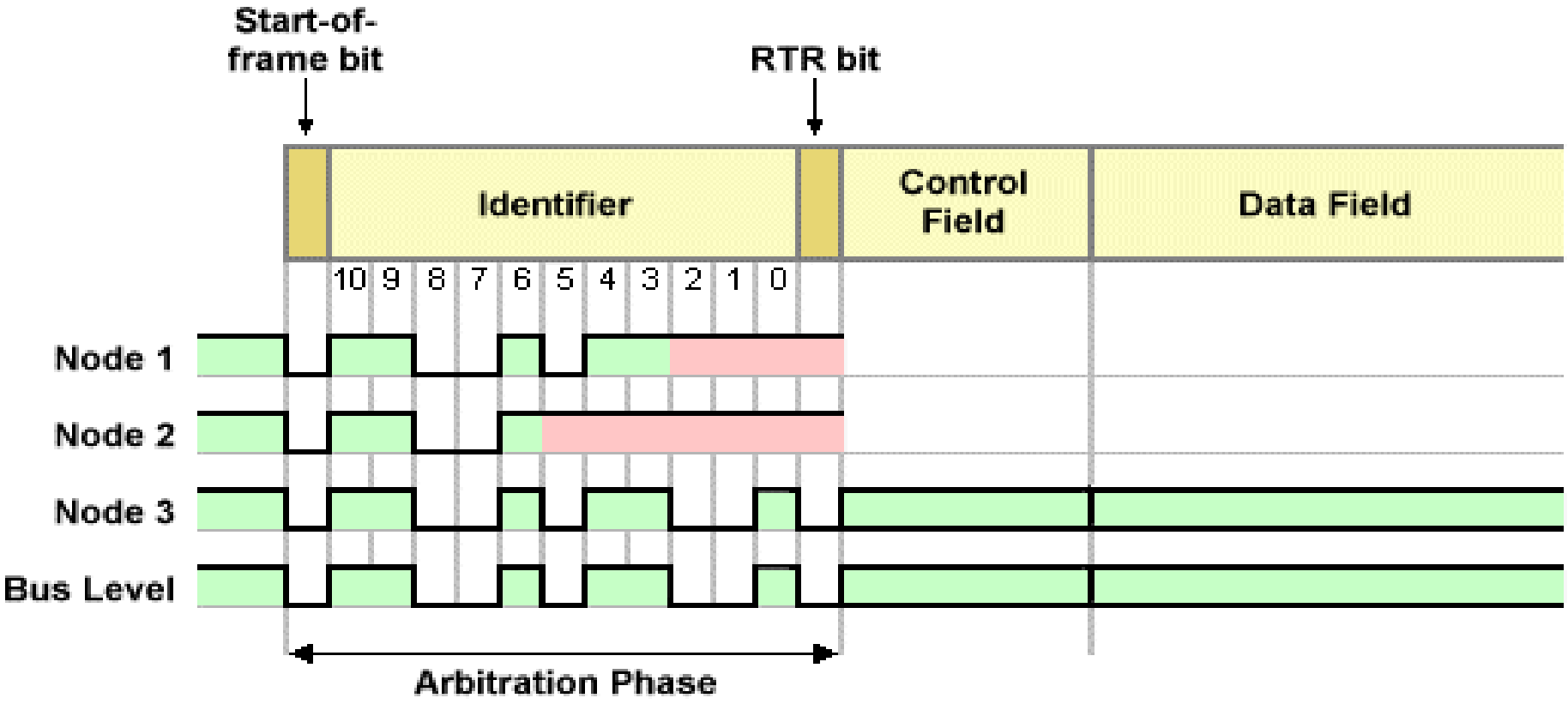
Referring Data Sheet MCP2515
SPI Based CAN Controller

STANDARD DATA&REMOTE FRAMES



BUS ARBITRATION

- The nodes are connected to the bus in a *wired-and* fashion: if just one node is driving the bus to a logical 0, then the whole bus is in that state regardless of the number of nodes transmitting a logical 1.



Conclusions

- Programming for any CAN Node needs Configuration(at sender).
  - identifier bits ; sent to indicate source/content/priority of message
  - RTR bit ; sent to indicate data/remote frame being transmitted
  - DLC bits ; to indicate no. of data bytes sent(DF)/or to be rcvd(RF)
  - Required Data bytes to be Written ;data bytes to be sents

- Programming for any CAN Node needs receive to process (at receiver)
  - identifier bits ; read by rcver to ingnore/consider source/content of msg
  - RTR bit ; read by receiver to know whether data/remote frame
  - DLC bits ; read by rcver for no of data bytes sent(DF)/to be sent(RF)
  - Required Data bytes to be Read ; data bytes read by rcver

- Remaining portions of Data/Remote/Error/Overload Frames will be managed by CAN controller .AND physical Signal by CAN transceiver.